Eenheid 3.1

Note Title 2016/03/30

Afgeleides van polinome en eksponensiaalfunksies.

Stewart p 172

1) Formules:

1.1 $\frac{d}{dx} c = 0$ c 'n konstant

Bv. $\frac{d}{dx}(2) = 0$

1.2 $\frac{d}{dx} x^n = nx^{n-1}$

$n = 1$

$$\frac{d}{dx}(x^1) = 1x^{1-1} = 1$$

Bv. $\frac{d}{dx}(x^2) = 2x^{2-1} = 2x$

- $\frac{d}{dt}\left(\frac{1}{\sqrt{t}}\right) = \frac{d}{dt}\left(t^{-\frac{1}{2}}\right) = -\frac{1}{2}t^{-\frac{1}{2}-1} = -\frac{1}{2}t^{-\frac{3}{2}}$
- $\frac{d}{du}\left(\frac{1}{u}\right) = \frac{d}{du}\left(u^{-1}\right) = -u^{-2}$
- $D\left(\frac{1}{v^3}\right) = D\left(v^{-3}\right) = -3v^{-4}$

1.3 $\frac{d}{dx}\left(e^{x}\right) = e^{x}$

Bv.
- $\frac{d}{du}\left(e^{u}\right) = e^{u}$
- $\frac{d}{dx}\left(e^{4}\right) = 0$

e^4 is 'n konstant

$e^4 \simeq (2.178)^4$

2. Reëls :

2.1 $$\frac{d}{dx} c f(x) = c \frac{d}{dx} f(x)$$

c 'n konstante

2.2 $$\frac{d}{dx}\left[f(x) + g(x)\right] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

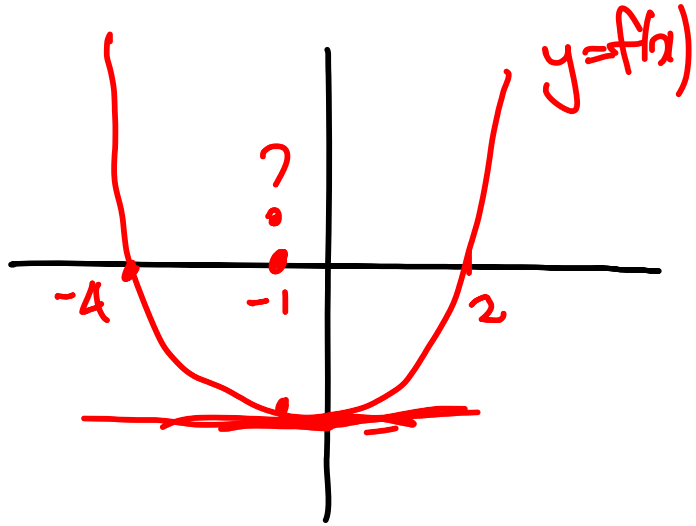
2.3 $$\frac{d}{dx}\left[f(x) - g(x)\right] = \frac{d}{dx} f(x) - \frac{d}{dx} g(x)$$

• Bepaal die x-waardes waar jy 'n horisontale raaklyn aan die grafiek van $f(x) = x^2+2x-8$ kan trek.

NB Horisontale raaklyn : $f'(x)=0$

$$f(x) = x^2+2x-8$$
$$= (x-2)(x+4)$$

$$f'(x) = 2x+2$$



$$f'(x)=0 \Rightarrow 2(x+1)=0 \Rightarrow x=-1$$

Bepaal $f'(2)$ as $y = 3e^{x} - \frac{4}{x}$ en skryf die vergelyking van die raaklyn neer as $x = 2$.

$f(x) = 3e^{x} - 4x^{-1}$ en $f(2) = y = 3e^{2} - 2$

$f'(x) = 3e^{x} + \frac{4}{x^{2}}$

$f'(2) = 3e^{2} + 1$

Vergelyking van raaklyn in $(2; f(2))$

$y - f(2) = f'(2)(x - 2)$

$$\Rightarrow y - (3e^2 - 2) = (3e^2 + 1)(x - 2) \; *$$

$$\Rightarrow y = (3e^2 + 1)x \;\; - 2(3e^2 + 1) + (3e^2 - 2)$$

$$\Rightarrow y = ?$$